



General Characteristics

101. Which one of the following statements concerning lanthanides elements is false

- (a) Lanthanides are separated from one another by ion exchange methods
- (b) The ionic radii of trivalent lanthanides steadily increase with increase in atomic number
- (c) All lanthanides are highly dense metals
- (d) Most typical oxidation of lanthanides is +3

102. Complex ion is shown by

- (a) *Ag* (b) *Au*
- (c) *Cu* (d) All of these

103. Which of the following transition metal is used as a catalyst

- (a) Nickel
- (b) Cobalt
- (c) Gold
- (d) Both (a) and (b)

104. For which of the following transition metal ion is shown highest magnetic moment having outer electronic configuration

- (a) $3d^1$ (b) $3d^8$
- (c) $3d^5$ (d) $3d^7$

105. The number of unpaired electrons is maximum in (Atomic no. : *Ti* = 22; *V* = 23; *Cr* = 24; *Fe* = 26)

(a) *Cr* (b) *Fe*

(c) *V* (d) *Sc*

106. Europium is

- (a) s-block element (b) p-block element
- (c) d-block element (d) f-block element

107. Which of the following elements is alloyed with copper to form brass

- (a) Lead (b) Silver
- (c) Zinc (d) Antimony

108. In which of the following metallic bond is strongest

- (a) *Fe* (b) *Sc*
- (c) *V* (d) *Cr*

109. Which of the following is a colourless ion

- (a) Cu^{+2} (b) Fe^{+3}
- (c) Ti^{+3} (d) Zn^{+2}

110. The substance used in cancer therapy is

- (a) *Rn* (b) *Ni*
- (c) *Fe* (d) *Co*

111. In solution of $AgNO_3$, if *Cu* is a solution become blue due to

- (a) Oxidation of *Ag*
- (b) Oxidation of *Cu*
- (c) Reduction of *Ag*
- (d) Reduction of *Cu*

112. Lanthanide for which + II and + III oxidation states are common

- (a) *La* (b) *Nd*



- (c) *Ce* (d) *Eu*
113. The number of unpaired electrons in Zn^{++} is
(a) 2 (b) 3
(c) 4 (d) 0
114. The first transition element is
(a) Chromium (b) Scandium
(c) Nickel (d) Copper
115. The electronic configuration (outermost) of Mn^{+2} ion (atomic no. of $Mn = 25$) in its ground state
(a) $3d^5 4s^0$ (b) $3d^4 4s^1$
(c) $3d^3 4s^2$ (d) $3d^2 4s^2 4p^2$
116. The highest magnetic moment is shown by the transition metal ion with the outer electronic configuration
(a) $3d^2$ (b) $3d^5$
(c) $3d^7$ (d) $3d^9$
117. Which has valency two
(a) *Al* (b) *Rb*
(c) *Cu* (d) *Ge*
118. All the following statements about the transitional elements are true except that
(a) All of the transitional elements are predominantly metallic
(b) In aqueous solution many of their simple ions are coloured
(c) Most of the transitional elements show pronounced catalytic activity
(d) Most of the transitional elements show only one valence state
119. Which of the following transition metals can have highest oxidation state
(a) *Cr* (b) *Co*
(c) *Ni* (d) *Cu*
120. Which of the following melts in boiling water
(a) Gun metal
(b) Wood's metal
(c) Monel metal
(d) Bell metal
(e) None of these

